SEPIDEH FARIVAR

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SUMMARY

Ph.D. candidate in Computer Science (expected May 2027) specializing in big data science, machine learning, and large language models for healthcare and public health. Builds end-to-end, reproducible systems on large, noisy administrative health data, including a MariaDB/Spark ETL pipeline over 59M Virginia All-Payer Claims Database (APCD) records from all payers, including Medicaid and Medicare, from which 325k pregnancy episodes were extracted and validated. Applies a broad modeling toolkit to maternal health outcomes (pregnancy loss, preterm birth, severe maternal morbidity): ordinal and multinomial regression, competing-risks survival, gradient boosting and Explainable Boosting Machines, sequence and FT-Transformers, and graph neural networks, with a strong emphasis on interpretability (SHAP, LIME), leakage-safe study design, and calibration for imbalanced clinical outcomes. Experienced in privacy-preserving AI: fine-tuning and serving open-weight LLMs entirely on-premise to convert coded claims into clinician-readable narratives, and designing a retrieval-augmented (RAG) clinical question-answering assistant on top of mined ICD-10-to-SMM association rules. Collaborates across computer science, nursing, and public policy, with a focus on clear documentation and research-to-deliverable execution.

RESEARCH INTERESTS

Health big data and informatics; claims-based risk prediction; maternal health outcomes and health disparities; interpretable and trustworthy machine learning; clinical NLP, LLMs, and retrieval-augmented generation; privacy-preserving AI for protected health data.

EDUCATION

Ph.D. in **Computer Science** — **University of Nevada, Las Vegas** — GPA: 3.9 — **2022 – May 2027 (expected)**
Research: Maternal health risk measurement pipeline using the Virginia All-Payer Claims Database. Advisor: Dr. Kazem Taghva

M.S. in **Computer Science** — **Mazandaran University, Babolsar**

B.Sc. in **Computer Science** — **Shahid Bahonar University of Kerman**

SKILLS

Programming:	Python, SQL, C++, Java, JavaScript, PHP, HTML5, CSS
Big Data & ETL:	Apache Spark (PySpark, Spark MLlib FPGrowth), Hadoop; large-scale ETL and rule-based validation pipelines (59M+ claims); MariaDB, MySQL
Machine Learning:	XGBoost, LightGBM, Logistic Regression, MLP, Bayesian classifiers, Explainable Boosting Machine (EBM), multinomial and ordinal regression, survival analysis with competing risks, latent class analysis, association-rule mining (FP-Growth, Apriori)
Deep Learning:	PyTorch, TensorFlow, Keras, Hugging Face Transformers; FT-Transformer, sequence Transformers; Graph Neural Networks (GCN, Heterogeneous Graph Transformer, PyTorch Geometric)
NLP & Large Language Models:	LLM fine-tuning (Med42-v2-8B, Llama 3), local deployment of open-weight LLMs (gpt-oss-120B, Llama 3.2, Llama 2, Falcon), GPT-3.5/4, BERT, DistilBERT, T5, MiniLM sentence embeddings; retrieval-augmented generation (RAG) design, embedding-based retrieval; prompt design, data-to-text generation, LLM-assisted annotation and NER, NLI-based hallucination checking, LLM-as-a-judge evaluation design
Interpretability & Evaluation:	SHAP (TreeExplainer, interaction values), LIME, surrogate decision trees, counterfactual analysis, probability calibration and threshold optimization, AUPRC for imbalanced outcomes, leakage control (landmark design, TRIPOD-aligned)
Healthcare Data:	All-Payer Claims Databases (including Medicaid and Medicare; enrollment/eligibility files), ICD-10-CM, CPT, pregnancy-episode construction, CDC severe maternal morbidity (SMM) indicators, CMQCC obstetric comorbidity scoring, gestational-age derivation
Secure Computing:	On-premise GPU computing (NVIDIA Quadro RTX 8000) for protected health data under data-use agreements; Linux; Git; Jupyter

Data Visualization:	Matplotlib, Seaborn, Plotly, Tableau
Web Development:	Laravel, Yii2, React.js, Mithril.js, Bootstrap, jQuery, Ajax
Other Tools:	OpenCV, IBM Rational Rose XDE, UML, Microsoft Visual Studio, Eclipse, IntelliJ IDEA, NetBeans, XAMPP, CuteFTP, Dreamweaver CC

RESEARCH EXPERIENCES

2026 **Designing a retrieval-augmented (RAG) clinical question-answering assistant** (planned). Nurses and physicians will be able to ask how a prenatal ICD-10 code in a patient's record relates to severe maternal morbidity (SMM); the system retrieves the relevant directional ICD-10→SMM association rules and clinical risk-factor evidence, and the locally fine-tuned clinical LLM generates a grounded, plain-language answer. Planned evaluation combines clinical domain experts with LLM-as-a-judge scoring. All components run on-premise under APCD data-use restrictions.

2026 **Developing a privacy-preserving LLM pipeline that turns longitudinal pregnancy-episode claims into fluent, clinician-readable English narratives** (in progress). Each Virginia APCD episode is converted from ICD-10/CPT codes into a chronological JSON timeline, rendered as deterministic template text, and rewritten into fluent English by a locally hosted gpt-oss-120B; the curated code-to-narrative pairs are then used to fine-tune an 8B clinical LLM (Med42-v2, Llama-3-based). All computation runs on-premise to comply with APCD data-use restrictions. Evaluated faithfulness of candidate models (rejected a 70B model that inserted clinical interpretation not supported by the data) and tested NLI-based hallucination detection. End goal: a clinician-facing chatbot that returns week-by-week events and a narrative summary for any requested episode. (Manuscript in preparation)

2026 **Leading a gestational-age-band-specific preterm birth study on 269,070 Virginia APCD live births** (30,807 preterm, 11.4%; in progress). Derived gestational age from delivery-claim Z3A codes and validated it against LMP-based dates; stratified preterm births into WHO bands (<28, 28–31, 32–33, 34–36 weeks) and spontaneous vs. medically indicated subtypes. Profiling prenatal risk factors and CDC severe maternal morbidity indicators per band with prevalence ratios and multinomial logistic regression adjusted for maternal age, payer (Medicaid vs. other), and year; comparing ordinal, competing-risks survival, sequence-Transformer, FT-Transformer, and Explainable Boosting Machine models under a landmark-based, leakage-controlled design. (Manuscript in preparation)

2026 **Conducted post-hoc interpretability analysis of an XGBoost Birth vs. Loss model on Virginia APCD pregnancy episodes.** Combined global SHAP (TreeExplainer) with exact SHAP interaction values to separate independent risk-factor effects from co-occurrence (e.g., severe preeclampsia), and explained six representative hardest and closest false negatives at the calibrated decision threshold with patient-level SHAP waterfall plots and LIME (10 random seeds for stability) over 25 clinical risk-factor groups. SHAP and LIME agreed in direction on 72.7% of 150 feature-level comparisons and 78.6% of present risk factors; findings were cross-checked with a surrogate decision tree and counterfactual analysis, showing that missed losses are driven mainly by sparse or protective risk evidence rather than model error on high-risk patients. (Manuscript in preparation)

2026 **Conducted distributed association-rule mining (Spark FPGrowth and Apriori) over 6M ICD-10 code records** across 291,761 pregnancy episodes and 16,855 ICD-10 codes, restricting antecedents to prenatal-period codes and consequents to SMM indicators at delivery or postpartum; produced 5,666 directional rules linking prenatal diagnoses to SMM and used an LLM (Llama 3.2) to generate clinician-oriented narrative interpretations. Built a rule-to-text dataset that serves as the knowledge base for the planned RAG question-answering assistant. (Basket Analysis and LLM-Assisted Interpretation of ICD-10 Association Rules in Maternal Health Data)

2025 **Designed an interpretability-first Birth vs. Loss risk-modeling pipeline on 241,076 Virginia APCD pregnancy episodes,** comparing XGBoost bag-of-codes (12,946 features) vs. an interpretable 285-feature categorical model and MiniLM sentence-embedding representations, and calibrating thresholds to target ~0.90 loss recall with detailed SHAP/LIME false-negative autopsies (From Codes to Concepts: A Methodological Case Study in Clinical Risk Modeling for Pregnancy Loss).

2025 **Developed Bayesian and DistilBERT-based classifiers over 14,127 ICD-10 codes** to distinguish birth vs. loss outcomes from Virginia APCD, targeting high recall on pregnancy loss while controlling precision. (Birth and Loss Classification Based on Maternal Insurance Claims Data)

2025 **Built a two-stage Transformer + rule-based framework** to automatically identify and number pregnancy episodes (prenatal, termination, postpartum) from longitudinal APCD claims for 397k patients, achieving high event-level F1 and >93% perfectly labeled pregnancies per patient. (AI-Assisted Pregnancy Episode Identification from All Payers Claim Dataset)

2025 Developed predictive models for maternal health outcomes using nursing-related APCD data, leveraging ICD code mapping, time-series structuring, and risk factor analysis across prenatal to postpartum stages. Engineered supervised

learning pipelines (Logistic Regression, XGBoost, MLP) to forecast pregnancy outcomes and analyzed feature importance to identify key maternal risk indicators.

2024 **Created a large-scale biomedical Named Entity Recognition (NER) dataset** by automatically annotating ClinicalTrials.gov eligibility criteria using Large Language Models (LLMs) for information extraction and text classification in healthcare.

2024 **Designed a scalable, MariaDB-based ETL and rule-based pipeline** to merge 59M Virginia APCD claims, extract and validate 325k pregnancy episodes, and label prenatal and postpartum care intervals, creating a reproducible dataset for downstream maternal risk modeling and policy analysis.

2024 Developed a novel "session-series" approach for exploratory analysis of gamblers' financial transactions to mine behavioral pattern data and contribute to responsible gambling initiatives.

2024 Computed the a-number for specific maximal curves over finite fields, employing the Cartier operator to derive a closed-form formula for this crucial invariant.

2024 Focused on algebraic geometry codes, including Goppa and quantum stabilizer codes, to enhance communication efficiency and reliability for high data rate transmission over noisy power line channels.

2023 Applied a GNN approach to detect fraudulent reviews in Amazon review products by leveraging advanced preprocessing, feature extraction (TF-IDF and BERT embeddings), graph construction, and an optimized GCN.

2022–2023 Focused on leveraging data analytics and advanced machine learning techniques to address the burgeoning issue of gambling addiction.

2015 Survey study on LDOF Algorithm using GPU to point trajectories by using OpenCV and C++.

2014 Research on different ways of point tracking using the Lucas-Kanade method with pyramids.

2013 Research on software engineering approach and database optimization; implemented design of a hotel booking system using IBM Rational Rose and SQL Server.

2012 Research on real-time workflow management in Arena; implemented simulation of a gas station.

2012 Research on optimizing game problem-solving methods using backtracking for solving Sudoku in C++.

2011 Research on the effect of the amount of night sleep time on students' intelligence using data mining in SPSS Clementine (80% training / 20% testing).

2010 Research on database optimization and software engineering: design, analysis, and evaluation of an information system for training registration and management (IBM Rational Rose and SQL Server).

TEACHING EXPERIENCE

~2 years **Instructor — DA 651: Managing Big Data & Web Databases** (graduate course), Data Analytics Program, Department of Computer Science, University of Nevada, Las Vegas

- Taught relational database design and SQL end to end: database creation, basic, conditional, join, and for-all (division) queries, constraints, indexing, transactions, and stored procedures.
- Covered web and modern data stacks: building database-driven web applications with PHP and MySQL/SQLite, accessing databases from Python, and NoSQL document-store databases, including when to choose SQL vs. NoSQL.
- Designed and graded about 10 hands-on SQL and database programming assignments per semester, using a rubric for solution correctness, syntax, and coding style and documentation.
- Held weekly office hours, guided students through query writing and debugging, and upheld academic-integrity and collaboration policies.

WORK EXPERIENCES

2022–Present **Graduate Research Assistant — UNLV, Department of Computer Science** (Advisor: Dr. Kazem Taghva)

Build and maintain the data engineering and machine learning pipeline for maternal health research on the Virginia All-Payer Claims Database (all payers, including Medicaid and Medicare): ETL, pregnancy-episode construction, risk-factor and SMM outcome definitions, predictive modeling, interpretability, local LLM fine-tuning, and retrieval-augmented question answering. Conduct research with the UNLV School of Nursing (including Dr. Vanderlaan) to translate model outputs into clinically meaningful findings for maternal care.

Summer 2024 **Research/Data Science Intern — UNLV, School of Public Policy** (Advisor: Prof. Benjamin Leffel)

- Standardized messy company/facility names and resolved duplicates in carbon-emissions datasets using Python (rule-based cleaning + fuzzy matching) to improve entity linking across sources.

- Used GPT (decoder-style LLM) via API to assist with text normalization/translation for noisy, multilingual records, improving consistency for downstream analysis.
- Prototyped QA and labeling workflows using BERT-based models (encoder-style) from Hugging Face to help classify emissions-related disclosures/responses and flag ambiguous cases for review.
- Built reusable, reproducible scripts and documentation so the cleaning pipeline could be rerun as new data arrived.

Improved data quality and entity matching across sources, helping enable the first global-scale analysis of corporate lobbying on climate change legislation.

2017–2021 **Sepehr System Technologists, Web Developer**

- Led the end-to-end redesign and development (front-end + back-end) of an online ticket-booking platform in Laravel and JavaScript, improving the user flow and checkout experience to support higher conversion.

2016–2017 **Zino Company, Back-end Developer**

- Developed and supported core back-end functionality in PHP, HTML, and CSS for a novel e-commerce startup and its smartphone applications, translating product ideas into features and improving the end-to-end order flow.

2015–2016 **Aryafa Company, Web Designer/Front-end Web Developer**

- Developed websites using the Yii 2.0 PHP Framework and the Mithril JavaScript Framework.

2015 **Hiva Afarinesh Company, SEO Developer/UI & UX Designer**

- Designed web pages using Bootstrap, Adobe Photoshop, and Adobe Illustrator CC.

2017 **Botanical website design:** online store for ornamental plants (HTML, CSS, Bootstrap, JavaScript, jQuery, Ajax, PHP, MySQL).

2017 **Advertising website design:** online store for promotional products (HTML, CSS, Bootstrap, JavaScript, jQuery, Ajax, PHP, MySQL).

PUBLICATIONS

- Nahed, P., Farivar, S., & Taghva, K. (2024). Automatic Annotation of ClinicalTrials.gov Entities Using Large Language Models. In *Proceedings of the International Congress on Information and Communication Technology* (pp. 431–439). Springer.
- Larni, M., Farivar, S., Puranik, P., Ghaharian, K., Golab, L., & Taghva, K. (2024). Exploratory analysis of gamblers' financial transactions to mine behavioral pattern data. In *Proceedings of the Third International Conference on Innovations in Computing Research (ICR'24)*. Springer.
- Mosallaei, B., Farivar, S., Ghanbari, F., & Nourozi, V. (2024). The a-number of $y^n = x^m + x$ over finite fields. arXiv. https://arxiv.org/abs/2404.08149
- Mosallaei, B., Ghanbari, F., Farivar, S., & Nourozi, V. (2024). Goppa codes: Key to high efficiency and reliability in communications. arXiv. https://arxiv.org/abs/2404.08132

UNDER PUBLICATION

- *Birth and Loss Classification Based on Maternal Insurance Claims Data* — accepted, Health Information Science and Systems (Springer), Nov. 2025.
- *AI-Assisted Pregnancy Episode Identification from All Payers Claim Dataset* — accepted, Health Information Science and Systems (Springer), Oct. 2026.
- *Transforming Insurance Claims into Maternal Care Insights* — submitted, IEEE BigData 2026, 9th Special Session on HealthCare Data.
- *Prevalence of Maternal Risk Factors across Antepartum and Intrapartum Claims* — submitted, PLOS One, Mar. 2026.
- *Validating the Classification of the Maternal Care Continuum in All-Payer Claims Databases* — submitted, Birth (Wiley), Oct. 2025.